IRRIGATION WATER MANAGEMENT · NOTE SP01

Soil–Plant–Water Relationships — Soil Properties and Soil-Water Constants

Course topics: IDE5.5 L10, CAE1.3 L11, MDC-IV L8

What the subject covers

Soil–plant–water relationships are the properties of soil and plants that govern how water moves, is held and is used. Soil stores water for roots to extract; water carries nutrients and is essential for growth; irrigation keeps a continuous supply of available moisture in the root zone.

- **For successful irrigation:** enough water of suitable quality; the right amount at the right time; a suitable method; and provision for removing excess water.
- **Good irrigation** gives higher yields, conserves land and water and keeps the soil productive. **Over-irrigation** causes erosion, leaching of fertilisers, waterlogging and salt build-up in the root zone.

Soil

Soil is a storehouse of plant nutrients, a home for bacteria, an anchorage for plants and a reservoir of water. In agriculture it means the thin upper layer of weathered rock penetrated by roots. Its physical properties decide how much water it holds in available form, and hence the frequency of irrigation and the capacity of the system.

- **Four components:** mineral matter, organic matter, water (liquid phase) and air (gaseous phase). A typical silt loam is 50% solids, 30% water and 20% air.
- **Properties that matter for irrigation:** infiltration, water-holding capacity, texture, structure, profile and depth of the water table.

Mechanical composition and texture

Fraction (USDA / ISSS)	Size
Gravel	2 mm to 2 cm
Fine earth (analysed)	< 2 mm
Sand	0.05 – 2.0 mm
Silt	0.002 – 0.05 mm
Clay	< 0.002 mm

- **Texture** is the relative proportion of sand, silt and clay. It controls water-holding capacity, permeability, infiltration and porosity.
- **Loam** — nearly equal sand, silt and clay; best for plant growth (medium-textured).
- **Sandy soils** — under 15% silt and clay; low water-holding capacity, good drainage and aeration (coarse-textured).
- **Clay soils** — at least 40% clay (the notes say "40% clay"); large surface area, high water and nutrient holding (fine-textured).

Structure and profile

- **Structure** is the arrangement of particles into aggregates, shaped by tillage, moisture at tillage and organic matter; it controls water and air movement, root penetration and nutrient supply. Granular, blocky and prismatic structures are favourable; platy structure impedes downward water movement; single-grained sands percolate rapidly.
- **Profile:** a vertical section from the surface to the parent material, made of distinct horizons differing in texture and structure; the pores hold soil air and soil water.
- **Rhizosphere:** the soil column occupied by roots. Only water in the root zone is available; as it decreases growth slows, so irrigate before yield is reduced.

Soil water

Soil water is the liquid phase held in the pores near the surface and in the root zone; it serves transpiration, physiological processes and evaporation. Excess water percolates down to groundwater or streams.

Kind	How it is held	Available to plants?
Hygroscopic water	Tightly, by adsorption	No
Capillary water	By surface tension; moves in all directions	Yes
Gravitational (free) water	Drains from large pores under gravity	Generally not retained for plant use

Movement	Meaning
Percolation	Downward movement through the profile under saturated or near-saturated conditions, governed by intrinsic permeability and hydraulic conductivity
Interflow	Lateral movement above a less permeable layer, reappearing lower down
Seepage	Vertical and lateral movement from canals, reservoirs, etc.

- **Intrinsic permeability** is a property of the porous medium alone — its ability to transmit air and water.
- **Hydraulic conductivity** is the rate of water flow through the soil, defined by **Darcy's law**: $v = K\,i$ (v = flow velocity, i = hydraulic gradient, K = hydraulic conductivity). It depends on the soil, the fluid, and salinity and alkalinity.

> **Correction.** The notes write Darcy's law as $v/i = k$ with k = intrinsic permeability. In Darcy's law the constant is the **hydraulic conductivity** K. Intrinsic permeability k is related to it by $K = k\,\rho\,g\,/\,\mu$, so it does not depend on the fluid.

Soil-moisture constants

Constant	Meaning
Soil-moisture tension	The force with which soil holds water, expressed as negative pressure, caused by adhesion and cohesion; the higher it is, the harder for plants to extract water
Saturation capacity	All pores full; the maximum water the soil can hold; tension ≈ zero
Field capacity	Moisture left after free water drains — 24–36 hours after irrigation; also called capillary capacity or field carrying capacity; about equal to the moisture equivalent and the one-third-atmosphere percentage
One-third atmosphere percentage	Moisture held at equilibrium with 1/3 atmosphere; closely approximates field capacity
Permanent wilting percentage (PWP)	Moisture at which plants wilt permanently, on a dry-weight basis; tension 7 to 32 atmospheres, commonly taken as 15 atmospheres
Available water	Field capacity - permanent wilting point; depends on texture and structure, tension, osmotic pressure and soil temperature

Soil (Table 8.2)	Available water-holding capacity (cm per m depth)
Fine sand	2–4
Sandy loam	4–11
Silt loam	6–13
Clay loam	10–18
Clay	16–30

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) lessons on soil–water–plant relations, evapotranspiration, irrigation efficiencies, and surface, sprinkler and drip irrigation. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Available water lies between

(a) saturation and field capacity
(b) field capacity and permanent wilting point
(c) PWP and hygroscopic coefficient
(d) zero and saturation

Q2. Permanent wilting point is commonly taken at a tension of

(a) 1/3 atm
(b) 1 atm
(c) 15 atm
(d) 31 atm

Q3. Field capacity corresponds closely to the moisture held at

(a) 0.1 atm
(b) 1/3 atm
(c) 15 atm
(d) 0 atm

Q4. Clay particles are smaller than

(a) 0.002 mm
(b) 0.02 mm
(c) 0.05 mm
(d) 2 mm

Q5. Water held by surface tension and available to plants is

(a) hygroscopic
(b) capillary
(c) gravitational
(d) combined

Q6. Available water-holding capacity of clay loam is about

(a) 2–4 cm/m
(b) 4–11 cm/m
(c) 10–18 cm/m
(d) 30–50 cm/m

Answers

Q1 — (b) field capacity and permanent wilting point
Q2 — (c) 15 atm
Q3 — (b) 1/3 atm
Q4 — (a) 0.002 mm
Q5 — (b) capillary
Q6 — (c) 10–18 cm/m